MIL-F-51214D(EA)
17 February 1989
SUPERSEDING
MIL-F-51214C(EA)
11 March 1988

MILITARY SPECIFICATION

FILTERS, GAS, 600, 1200, 2500 AND 5000 CFM; C22R1, C32R1, C29R1 AND C23R1

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of gas filters.

1.2 Classification. The gas filters shall be furnished in the following types as specified:

Type	Capacity (cubic feet per minute)
C22R1	600
C32R1	1200
C29R1	2500
C23R1	5000

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 4240

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants.

STANDARDS

FEDERAL

FED-STD-101 - Test Procedures for Packaging Materials.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection Attributes.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

TDPL 5-19-1851 - Filter, Gas, 2500 CFM, C29R1.
TDPL 5-19-1863 - Filter, Gas, 600 CFM, C22R1.
TDPL 5-19-1866 - Filter, Gas, 1200 CFM, C32R1.
TDPL 5-19-1868 - Filter, Gas, 5000 CFM, C23R1.

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SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-1851 - Filter, Gas, 2500 CFM, C29R1.
P5-19-1863 - Filter, Gas, 600 CFM, C22R1.
P5-19-1866 - Filter, Gas, 1200 CFM, C32R1.
P5-19-1868 - Filter, Gas, 5000 CFM, C23R1.

(Copies of specifications, standards, drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standard

E84 - Surface Burning Characteristics of Building Materials.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORIES, INC.

UL Standards

UL723 - Test Method for Fire Hazard Classification of Building Materials.

(Application for copies should be addressed to Underwriters Laboratories, Inc., 207 E. Ohio Street, Chicago, IL 60611).

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 **Materials.** All materials cited on the applicable TDPL 5-19-1851, TDPL 5-19-1863, TDPL 5-19-1866 or TDPL 5-19-1868 and on the subsidiary drawings shall conform to the specification listed thereon or to the specific characteristics set forth on the drawings and in this specification.

3.1.2 **Components.** All components of the filters shall conform to the specifications and drawings on the applicable TDPL.

3.2 **Assembly.** The filters shall be assembled and marked in accordance with the following drawings:

Type	Drawing number
C22R1	5-19-1863
C32R1	5-19-1866
C29R1	5-19-1851
C23R1	5-19-1868

3.3 **Plywood.** The plywood used in the construction of the filters shall have a flame spread classification of 25 or less when tested as specified in ASTM Standard E84 or UL723.

3.4 **Leakage.** The filter shall not leak when a concentration of 1000 parts per million (ppm) of dichlorodifluoromethane refrigerant (R-12) conforming to type 12 of BB-F-1421, in air is introduced at the influent side of the filter at 1/5 the rated airflow of the filter in accordance with 4.4.4.1. A filter leak shall be defined as the detection of 1 ppm or more of R-12 in the air stream effluent of the filter within 2 minutes after the introduction of R-12 at the influent side.

3.5 **Airflow resistance.** The airflow resistance of the 600 cubic feet per minute (cfm) filter shall not exceed 1.25 inches of water, for the 1200 and 2500 cfm filters it shall not exceed 1.50 inches of water and for the 5000 cfm filter it shall not exceed 2.00 inches of water at their rated airflows when tested as specified in 4.4.4.2.

3.6 **Airflow resistance and dimethylmethylphosphonate (DMMP) value after rough handling.** After the filter has been rough handled with the filter modules in the horizontal position for 30 minutes as specified in 4.4.4.3, the airflow resistance shall meet the requirements of 3.5 and the DMMP gas life shall be not less than 35 minutes when tested in accordance with 4.4.4.4 and the following test parameters:

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Airflow rate	C22R1	600 ± 30 cfm
	C32R1	1200 ± 60 cfm
Airstream temperature		110 ± 5°F
DMMP influent concentration		5.0 ± 0.1 mg/1
DMMP break concentration		0.04 ± ug/1

3.7 **Preproduction.** Prior to the start of regular production, a preproduction sample of gas filters of the same type shall be produced in accordance with this specification for examination and test (see 4.2).

3.8 **Workmanship.** The filter shall be free from foreign matter, such as dirt, grease, oil or viscous material or damage such as cracked or distorted wooden sections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection.

4.3.1 Sample. Prior to the start of regular production, a first article sample of filters shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production. First article sample sizes are as follows (see 6.2):

Four C22R1 gas filters.
Two C32R1 gas filters.
Two C29R1 gas filters.
Two C23R1 gas filters.

The sample of filters shall be packaged in accordance with their respective SPIs (see 5.1), unless interplant shipment is required.

4.3.2 Inspection procedure.

4.3.2.1 For examination and tests. As determined by the Government, the first article sample shall be examined and tested for any or all of the requirements of the applicable drawings and this specification.

4.3.2.2 For non-destructive tests. Each packaged filter in the first article sample shall be tested for leakage (4.4.4.1), airflow resistance (4.4.4.2), and unit pack leakage (4.4.4.5).

4.3.2.3 For destructive tests. Two C22R1 or two C32R1 filters shall be selected from the first article sample and tested for airflow resistance and DMMP gas life after rough handling (4.4.4.4). The first article sample of packaged filters shall be tested for heat seal seam strength (4.4.4.6) after the non-destructive tests have been completed.

4.3.2.4 Rejection criteria. The first article sample shall be rejected if it fails to meet any of the requirements of the applicable drawings and this specification. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the filters of one type produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of filter modules shall be represented in any one lot of finished filters.

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4.4.2 Sampling.

4.4.2.1 For examination and test. Sampling of the lot for non-destructive tests shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Two C22R1 or two C32R1 filters shall be tested for airflow resistance and DMMP gas life after rough handling.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample filters shall be examined in accordance with the classification of defects (4.4.3.5) and MIL-STD-105.

4.4.3.2 For non-destructive test. Each sample (4.4.2) packaged filter shall be tested for unit pack leakage (4.4.4.5). Any assembly found defective shall be rejected.

4.4.3.3 For destructive tests. Two C22R1 or two C32R1 filters from the sample (4.4.2) shall be tested for airflow resistance and DMMP gas life after rough handling (4.4.4.4). Each filter must meet the minimum requirement for the lot to be acceptable. The sample of packaged filters shall be tested for heat seal seam strength (4.4.4.6), after the non-destructive tests have been completed, using inspection level S-4 and an AQL of 2.5 percent.

4.4.3.4 For critical defects. Each filter in the lot shall be tested for the critical characteristics listed in the classification of defects. Any filter failing to meet the requirements of 3.4 and 3.5 shall be rejected.

4.4.3.5 Classification of defects.

(a) Filters, gas, 600, 1200, 2500, 5000 cfm (Drawing 5-19-1863, 5-19-1866, 5-19-1851 and 5-19-1868).

Categories	Defects	Acceptance standards
Critical:	None defined	
1	Leakage	4.4.4.1
Major:	100% inspection	
101	Airflow resistance	4.4.4.2

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AQL 1.0 percent defective

102	Component missing
103	Sealant incorrectly applied or missing
104	Component incorrectly assembled
105	Marking incorrect
106	Workmanship (3.8)

Minor: AQL 2.5 percent defective

201	Marking illegible

(b) Packaging inspection (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective

101	Items not completely clean/dry
102	Unit pack assembly incorrect
103	End panels missing or incorrect
104	Cushioning missing or incorrect
105	Unit pack container and closure incorrect
106	Trays missing or incorrect
107	Barrier missing or incorrect
108	Strapping incorrect
109	Marking missing, incorrect or illegible
110	Humidity indicator missing
111	Unit pack bag and closure incorrect

4.4.4 Tests. Tests shall be conducted as follows.

4.4.4.1 Leakage. The filter shall be tested for leakage as follows:

(1) Connect the filter to a source of forced air. Also, place an efficient mixing chamber at the effluent side of the filter. The filter shall be without transit and/or packaging material.

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(2) Regulate the flow of air from the blower to 1/5 the rated airflow of the filter being tested. Measurement of the airflow is determined by the orifice and draft gage.

(3) Introduce R-12 into the intake of the blower; monitor and maintain a concentration of 1000 ppm by volume of R-12 in air at 1/5 rated airflow on the influent side of the filter for the specified period of time.

(4) Determine the leakage using a suitable leak detector (see 6.4) for sampling and detecting R-12 in the mixing chamber at the effluent side of the filter.

(5) Purge the filter by passing fresh air in the reverse direction of that indicated in (3) above through the filter at its rated airflow for 1 minute.

(6) Keep the exposure of the filter to air to a minimum. The airflow time for leakage should not exceed 3 minutes.

4.4.4.2 Airflow resistance. The pressure drop across the filter shall be determined at the rated airflow, evenly distributed over the face of the filter. The up and down stream static pressure measuring tubes shall be as close as possible to the filter, and shall not be on a section of duct having a changing cross sectional area.

4.4.4.3 Rough handling. Sample filters shall be rough handled in accordance with MIL-STD-810, method 514, procedure XI, except part I and with the following exceptions to part 2:

(1) The inlet and outlet openings of each filter shall be covered with 1/4 inch thick plywood flat panel (to prevent damage to exposed surface), nailed on as specified in applicable SPI (see section 5). The filter shall be without transit case or packaging material.

(2) The time shall be 30 minutes with the filter remaining on one side throughout the test.

(3) Constrain the filter to a horizontal motion of not more than 1/2 inch.

4.4.4.4 Airflow resistance and dimethylmethylphosphonate (DMMP) value after rough handling. After the filter has been rough handled as specified in 4.4.4.3, the filter shall be tested for airflow resistance in accordance with 4.4.4.2, and shall be tested for DMMP value in an apparatus which establishes the following test conditions:

(a) Preheat sample filter 120 ± 10°F for a minimum of 30 minutes.

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(b) While in the bypass mode adjust the controls until rated airflow is reached and the temperature is 110 ± 5°F. Generate DMMP until the influent concentration detector (see 6.6) indicates 5.0 ± 0.1 milligrams DMMP per liter air.

(c) Insert preheated sample filter using applicable test fixture if necessary.

(d) Switch to test mode and start timer. Make final adjustments to airflow, temperature, and DMMP concentration.

(e) Stop the test and record the time when the effluent concentration detector indicates that the break concentration has been reached.

4.4.4.5 Unit pack bag leakage. The leakage of the unit pack bag containing the filter shall be determined by subjecting the sample to the pneumatic pressure technique, method 5009.2 of FED. TEST METHOD STD. NO. 101. The test pressure shall be 1 inch H_2O. The sides and top of the unit box, and the strapping, shall be removed from around the unit bag prior to conducting the test.

4.4.4.6 Heat seal seam. The heat seal seam strength of the unit pack bag shall be determined by subjecting the sample to FED. TEST METHOD STD. NO. 101, method 2024, except that the static load shall be 480 ounces ± 16 ounces, and a 5 percent reduction shall be permitted when room temperature in test area exceeds 90°F. The test sample shall be taken from one end of the heat sealed closure in order to permit resealing of the bag after testing.

5. PACKAGING

5.1 Interplant shipment (see 6.7). Military packaging as specified in 5.2 is not applicable for interplant shipments. The filters shall be packaged to provide adequate protection from physical damage, and marked to identify the filters, and provide any other required data during shipment from the source to the first receiving activity.

5.2 Military packaging. For all shipments, other than interplant shipments (see 6.7), packaging (preservation, unit packing, packing and marking) shall be as specified in SPI P5-19-1851, P5-19-1863, P5-19-1866, or P5-19-1868, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gas filters covered by this specification are components of filter units used for protecting fixed installations against toxic gases and particulate matter. The following is a list of gas filters and their respective assemblies:

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Type	Assembly
C22R1	Filter, Gas-Particulate, 600 CFM, M14A1.
C32R1	Filter, Gas-Particulate, 1200 CFM, M15A1.
C29R1	Filter, Gas-Particulate, 2500 CFM, M16A1.
C23R1	Filter, Gas-Particulate, 5000 CFM, M17A1.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of gas filter required (see 1.2).
(c) Shelf life code.
(d) Type of packaging required.
(e) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 DMMP influent concentration detector. The Beckman model 400 hydrocarbon analyzer has been found to be a suitable influent DMMP concentration detector.

6.4 DMMP effluent concentration detector. The Meloy Laboratories phosphorous analyzer, Model SA-160 has been found to be a suitable DMMP effluent concentration detector.

6.5 Subject term (key word) listing.

Airflow resistance
Filter, gas

6.6 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 214. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 Interplant shipment. Packaging and marking for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime

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contractor, or between contractors and subcontractors, or from a vendor or contractor to a military arsenal, plant, or other activity for evaluation, immediate use, or further processing as specified in the applicable contract.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - EA

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51214D(EA)	FILTERS, GAS, 600, 1200, 2500 AND 5000 CFM; C22R1, C32R1, C29R1 AND C23R1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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